

CHAPTER 1

INTRODUCTION

1.1 General Introduction

The Smart Wearable Glove for Sign Language Translation is developed to help deaf and mute people communicate easily with others. The system uses flex sensors attached to a glove to detect finger bending and hand gestures. When the user performs a sign, the resistance of the flex sensors changes according to the finger movement. These sensor values are read and processed by the Arduino microcontroller. The processed data is then transmitted wirelessly using the HC-05 Bluetooth module. The received gesture is converted into readable text or speech output on a smartphone. The project provides a portable, low-cost, and real-time communication solution. It also demonstrates the application of embedded systems, sensors, and wireless communication technologies in assistive devices.

1.2 Motivation

Communication is an essential part of human life, but deaf and mute people often face difficulties in interacting with others because they mainly depend on sign language for communication. Existing communication methods often require human interpreters, which may not always be available. This created the motivation to develop a smart and portable system that can translate sign language gestures into understandable text or speech in real time.

Recent advancements in wearable devices, IoT, sensors, and wireless communication technologies further inspired this project. Flex sensors can accurately detect finger bending and hand movements, while the Arduino microcontroller processes the sensor data efficiently. The HC-05 Bluetooth module enables wireless communication between the glove and a smartphone or display device. The project focuses on developing a low-cost, compact, and user-friendly assistive device that improves communication and independence for deaf and mute individuals. Moreover, this system demonstrates the practical application of embedded systems and intelligent human-machine interaction technologies in solving real-world social problems.

1.3 Background

The advancement of technology has significantly improved communication systems and assistive devices for especially abled individuals. Deaf and mute people mainly depend on sign language to communicate, but most people are not trained to understand these gestures. This creates a major communication barrier in daily life, especially in educational institutions, hospitals, workplaces, and public areas. To overcome this problem, researchers have focused on developing smart systems that can recognize hand gestures and convert them into readable text or speech. Wearable technology combined with embedded systems has emerged as an effective solution for sign language translation.

Recent developments in sensors, microcontrollers, and wireless communication technologies have made gesture recognition systems more accurate and efficient. Flex sensors are widely used to detect finger bending because their resistance changes according to the amount of bend. Microcontrollers such as Arduino can process these sensor signals and identify specific gestures. Wireless modules like HC-05 Bluetooth enable real-time transmission of gesture data to smartphones. These technologies have encouraged the development of smart wearable gloves that provide portable, low-cost, and real-time communication assistance for deaf and mute people. The proposed project is based on these advancements and aims to develop a practical and user-friendly sign language translation system.

1.4 Problem Statement

Communication between deaf and dumb individuals is difficult because sign language is not widely understood. This project proposes a low-cost Arduino-based sensor glove that converts hand gestures into text for real-time communication.

1.5 About the Project

The project titled “Smart Wearable Glove for Sign Language Translation” is designed to help deaf and mute people communicate easily with others by converting hand gestures into understandable text or speech. The system uses flex sensors mounted on a wearable glove to detect finger bending and hand movements. Whenever the user performs a sign language gesture, the flex sensors produce corresponding electrical signals based on the bending of the fingers. These sensor values are processed by the Arduino microcontroller, which identifies the gesture according to predefined conditions.

The processed data is transmitted wirelessly to a smartphone using the HC-05 Bluetooth module. The output is then displayed as text or converted into speech, enabling real-time communication between especially abled individuals and normal users. The project mainly focuses on developing a portable, low-cost, and user-friendly assistive device using embedded systems and wireless communication technologies.

The proposed system demonstrates the practical application of sensors, Arduino programming, and Bluetooth communication in gesture recognition systems. It reduces communication barriers faced by deaf and mute people and provides an efficient human-machine interaction platform. The project can be further enhanced by integrating machine learning, artificial intelligence, and mobile applications for more accurate and advanced sign language recognition.

1.6 Objectives of the Project

1. To design a wearable sensor glove that captures hand gestures using flex sensors.
2. To recognize sign language gestures using Arduino UNO.
3. To convert recognized gestures into readable text on mobile display in real time.

1.8 Working Principle

The Smart Wearable Glove for Sign Language Translation works by detecting finger movements using flex sensors attached to the glove. When the user bends their fingers, the resistance of the flex sensors changes according to the bending angle. These resistance changes are converted into voltage signals using a voltage divider circuit with a 10k Ω resistor. The Arduino microcontroller reads these analog voltage values through its input pins. Different hand gestures produce different combinations of sensor values. The Arduino continuously compares these values with predefined gesture patterns stored in the program. Once a gesture is recognized, the corresponding text or message is generated. The HC-05 Bluetooth module then transmits the data wirelessly to a smartphone or display device. The received output is displayed as readable text or converted into speech. Thus, the system enables real-time communication for deaf and mute individuals using hand gestures.

1.9 Advantages

1. Helps deaf and mute people communicate easily with others.
2. Converts sign language gestures into understandable text or speech.
3. Provides real-time communication using wearable technology.
4. Portable, lightweight, and easy to use in daily life.
5. Wireless communication is achieved using the HC-05 Bluetooth module.
6. Reduces dependency on human interpreters for communication.
7. Uses simple and efficient embedded system technology.
8. Improves social interaction and independence of especially abled individuals.

1.10 Disadvantages

1. The system accuracy depends on proper gesture calibration.
2. Limited number of gestures can be recognized in the basic system.
3. Different users may produce slight variations in gestures, affecting accuracy.
4. Bluetooth communication range is limited.
5. Practical implementation may require advanced programming and testing for higher accuracy.

1.11 Applications

1. Used for communication assistance for deaf and mute people.
2. Helps in real-time sign language translation into text or speech.
3. Used in hospitals and emergency situations for easy communication.
4. Applied in smart healthcare and assistive wearable devices.
5. Used in human-computer interaction systems.
6. Helpful in public places where sign language interpreters are unavailable.
7. Useful for research in gesture recognition and embedded systems.
8. Can be integrated into mobile applications for advanced communication support.

CHAPTER 2

LITERATURE SURVEY

1. IoT-Enabled Assistive Glove for Real-Time Sign Language Translation (2025)

M. A. Rahman, S. Islam, and T. Hasan proposed an IoT-enabled smart glove using flex sensors, MPU-6050, Arduino Nano, and Bluetooth communication for real-time sign language translation. A Random Forest machine learning algorithm achieved nearly 97% gesture classification accuracy, improving communication efficiency between deaf users and normal speakers.

2. Real-Time Sign Language to Speech Conversion Glove Using Arduino and Wearable Sensors (2025)

F. Khan and R. S. Chahar developed a low-cost wearable glove system using Arduino, flex sensors, and an accelerometer for converting sign language gestures into speech. The system used Bluetooth communication for real-time data transmission and focused on portability, affordability, and assistive communication technology.

3. Sign Language Detection and Translation Using Smart Glove (2024)

S. Maharjan, S. Shrestha, and S. Fernando designed a smart glove system using five flex sensors and an MPU-6050 sensor for sign language recognition. The proposed system used Long Short-Term Memory (LSTM) based gesture recognition to translate gestures into text and speech in real time.

4. Smart Glove for Malayalam Sign Language Recognition and Audio Output (2024)

Muralikrishna K. S., Nikhitha Prakasan, Haritha T. K., Asha J. George, and Rini T. Paul developed a wearable glove system for Malayalam sign language recognition using flex sensors, MPU sensor, Arduino Nano, and Wi-Fi/Bluetooth communication. The system converted gestures into text and audio output for assisting deaf and mute users.

5. Wearable Sensor-Based Hand Gesture Recognition System for Speech Impaired People (2023)

Researchers proposed a wearable sensor glove integrated with an Android application for real-time gesture recognition and communication assistance. The system displayed translated text and generated speech output to improve interaction between deaf users and normal individuals.

CHAPTER:3

SYSTEM REQUIREMENTS AND SPECIFICATION

3.1 Hardware Requirements

The hardware components required for the proposed Smart Wearable Glove for Sign Language Translation system are listed below:

Table 3.1.1: Hardware Requirements

Sl. No	Component	Specification
1	Arduino Uno	ATmega328P Microcontroller
2	Flex Sensors	2.2-inch Flex Sensors
3	Bluetooth Module	HC-05 Bluetooth Module
4	Resistors	10k Ω Resistors
5	Power Supply	5V DC Supply
6	Connecting Wires	Jumper Wires
7	Glove	Wearable Hand Glove
8	Smartphone	Android Mobile Device

3.2 Software Requirements

The software tools used for developing and testing the system are:

Table3.2.1: Software requirement

Sl. No	Software	Purpose
1	Arduino IDE	Programming Arduino Uno
2	Embedded C	Writing program code
3	Bluetooth Terminal App	Displaying output text
4	Windows OS	System development platform

3.3 Functional Requirements

1. **Gesture Detection:** The flex sensors should detect finger bending and hand movements accurately. Different gestures should produce different sensor output values.
2. **Sensor Data Processing:** The Arduino should continuously read and process sensor values. The controller should compare sensor data with predefined gestures.
3. **Gesture Recognition:** The system should identify specific hand gestures correctly. Each gesture should correspond to a particular message or command.
4. **Wireless Communication:** The HC-05 module should transmit data wirelessly to the smartphone. Communication should occur with minimum delay and data loss.
5. **Text Output:** The recognized gesture should be converted into text output. The output should be displayed clearly on the connected device.
6. **User-Friendly Operation:** The glove should be lightweight and comfortable to wear. The system should be simple and easy to operate.
7. **Low Power Consumption:** The system should consume less electrical power during operation. It should support longer operation using portable power sources.

3.4 Design Constraints

1. **Sensor Accuracy Limitation:** Flex sensor readings may vary due to improper finger bending. Sensor aging can reduce gesture recognition accuracy over time.
2. **Limited Number of Gestures:** Arduino memory limits the number of stored gesture patterns. Only predefined gestures can be recognized in the basic system.
3. **Bluetooth Range Limitation:** The HC-05 module supports only short-range communication. Communication quality decreases beyond the specified range.
4. **Power Supply Requirement:** The system requires a stable 5V power supply for operation. Power interruption can affect system performance.
5. **Flex Sensor Durability:** Continuous bending may reduce sensor sensitivity over time. Long-term usage can increase maintenance requirements.
6. **Environmental Noise:** Electrical noise may affect analog sensor readings. Proper calibration is required to minimize errors.
7. **System Calibration:** The glove requires proper calibration before use. Incorrect calibration may lead to false gesture detection.

3.5 System Specifications

Table 3.5.1: System Specifications

Parameter	Specification
Microcontroller	Arduino Uno
Sensor Type	Flex Sensor
Communication	Bluetooth HC-05
Operating Voltage	5V
Programming Language	Embedded C
Wireless Range	Approximately 10 meters
Output Type	Text/Speech
Platform	Android Smartphone
Gesture Detection	Real-Time
System Type	Wearable Assistive Device

CHAPTER 4

PROPOSED METHODOLOGY

4.1 Selection of Sensors

Flex sensors are selected as the main sensing elements for detecting finger bending and hand gestures. These sensors are mounted on the glove and produce varying resistance values according to finger movement.

4.2 Design of Wearable Glove

A wearable glove is designed to hold the flex sensors properly on the fingers. The glove ensures comfortable usage and accurate gesture detection during hand movement.

4.3 Signal Conversion Using Voltage Divider

The flex sensors are connected with 10k Ω resistors to form voltage divider circuits. This arrangement converts resistance changes into corresponding voltage signals for Arduino processing.

4.4 Arduino-Based Processing

Arduino Uno microcontroller is used to read analog voltage values from the flex sensors. The controller processes the sensor data and compares it with predefined gesture patterns stored in the program.

4.5 Gesture Recognition

Different combinations of sensor values represent different hand gestures. The Arduino identifies the performed gesture by comparing real-time sensor readings with programmed threshold values.

4.6 Bluetooth Communication

The HC-05 Bluetooth module is used for wireless communication between the glove and the smartphone. The recognized gesture data is transmitted wirelessly through serial communication.

4.7 Text and Speech Generation

After recognizing the gesture, the corresponding message is generated as text or speech output. The translated output is displayed on a smartphone or converted into voice for easy communication.

4.8 Real-Time System Operation

The complete system operates continuously in real time for fast communication support. Sensor detection, processing, and output generation occur with minimum delay.

4.9 Testing and Validation

The developed system is tested using different hand gestures to verify accuracy and performance. Sensor readings and output responses are analysed to ensure reliable gesture recognition.

4.10 Flow of work

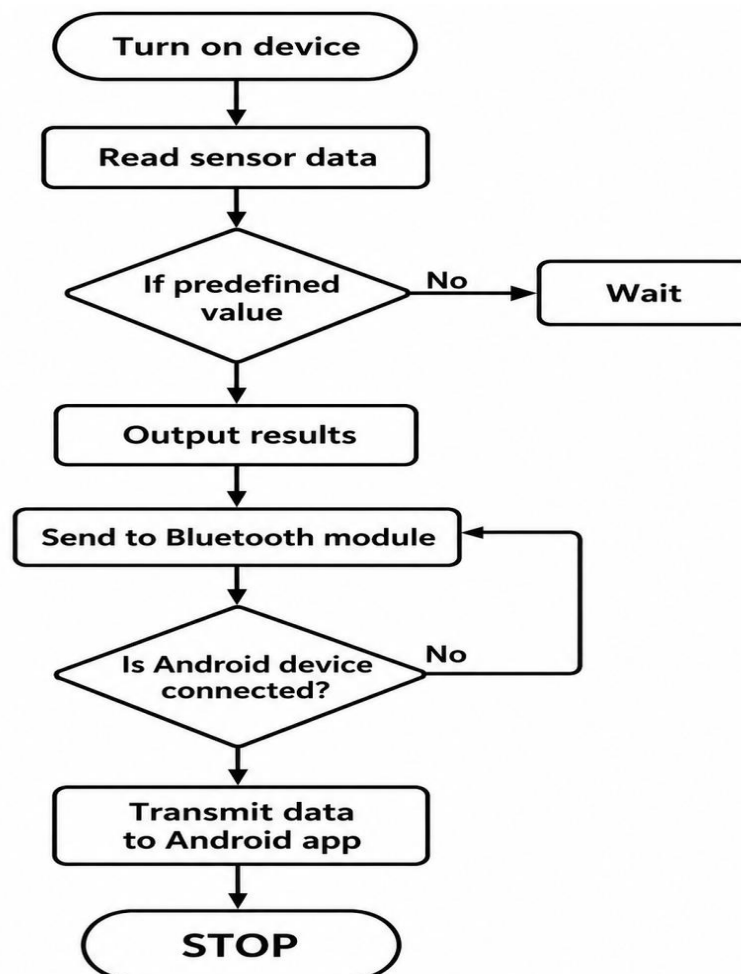


Figure 4.10: Flowchart of the project

CHAPTER 5

DESIGN OF THE PROJECT

5.1 Block Diagram

The block diagram of the Smart Wearable Glove for Sign Language Translation explains the complete working process of the system. Initially, the user performs different hand gestures using the wearable glove. The flex sensors attached to the glove detect the bending of fingers and produce varying resistance values according to finger movement. These sensor outputs are given to the Arduino UNO microcontroller, which reads and processes the analog signals.

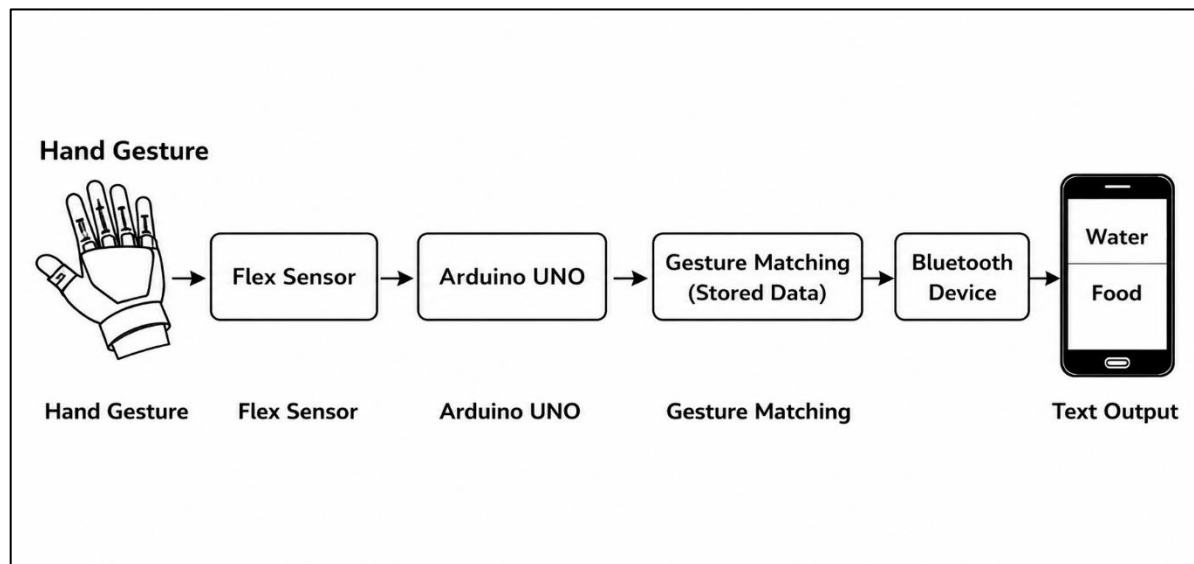


Figure5.1.1: Block Diagram of the Project

The processed sensor values are then compared with predefined gesture patterns stored in the Arduino memory through the gesture matching process. If the performed gesture matches the stored data, the corresponding command or message is selected. The recognized data is then sent to the HC-05 Bluetooth module, which transmits the information wirelessly to an Android smartphone or application. Finally, the translated output is displayed as readable text on the mobile screen, enabling real-time communication for deaf and mute individuals.

5.2 Components Used

5.2.1 Arduino UNO

Arduino UNO acts as the main controller of the entire system. It continuously reads the analog voltage values produced by the flex sensors through its analog input pins. The Arduino processes these sensor readings and compares them with predefined gesture values stored in the program memory. Once a gesture is identified, Arduino generates the corresponding text command and sends it to the Bluetooth module for wireless transmission.

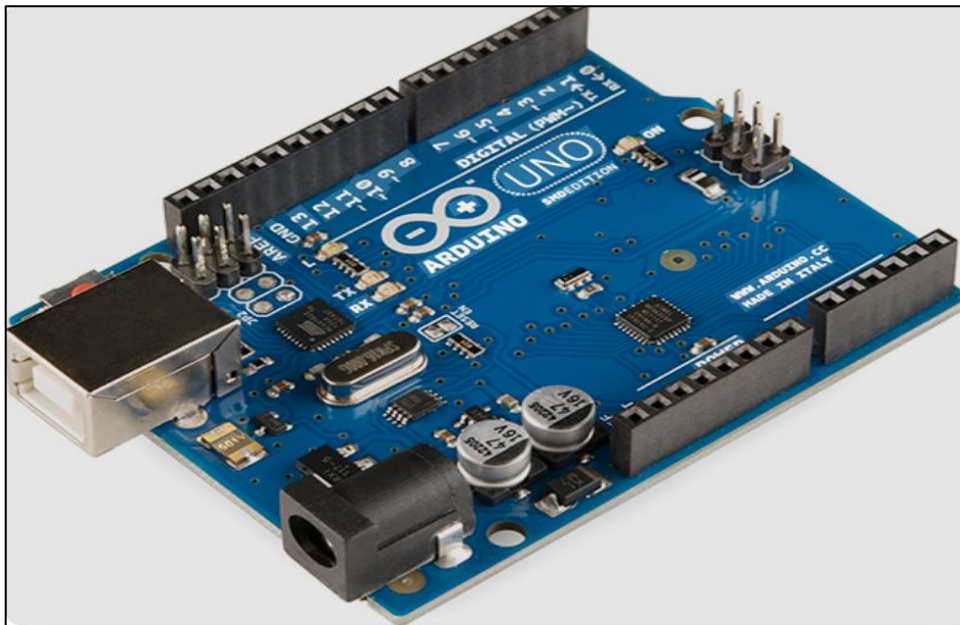


Figure 5.2.1.1: Arduino UNO

5.2.2 Flex Sensors

A flex sensor is a type of variable resistor whose resistance changes when it bends. When the sensor remains straight, it has a normal resistance value, and when it bends, the resistance increases. In the Smart Wearable Glove project, flex sensors are attached to the fingers of the glove to detect finger movements and hand gestures.

The flex sensor is connected with a $10\text{k}\Omega$ resistor to form a voltage divider circuit. Due to finger bending, the resistance of the sensor changes, which produces different voltage values at the output. These changing analog voltage signals are sent to the Arduino UNO through its analog input pins. The amount of voltage variation depends on the bending angle of the finger.

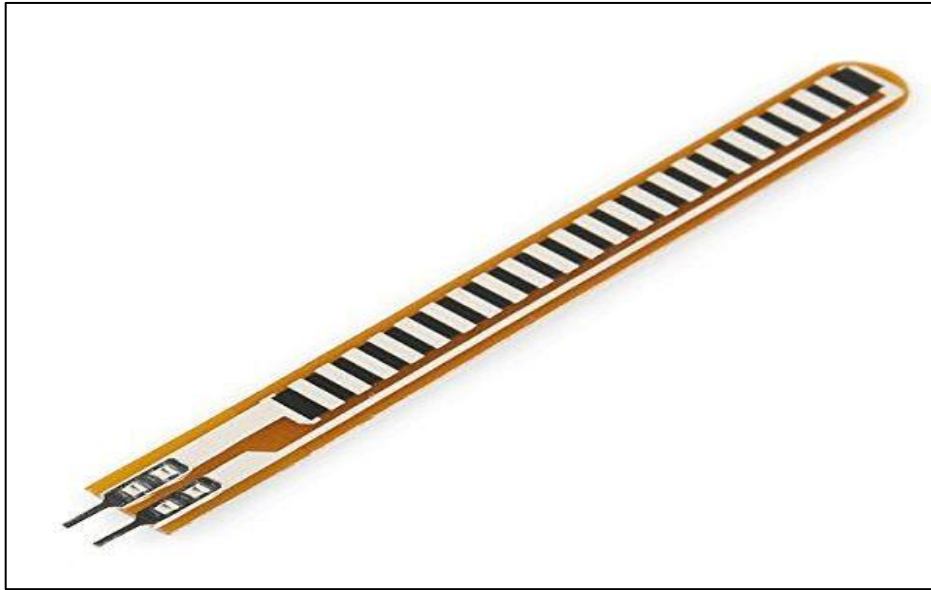


Figure5.2.2.1: Flex Senor

The Arduino continuously reads and processes these sensor values using its Analog-to-Digital Converter (ADC). Different finger positions generate different sensor readings, which are compared with predefined gesture values stored in the program. Based on these readings, the Arduino identifies the performed gesture and sends the corresponding message through the Bluetooth module for text or speech output.

5.2.3 Bluetooth HC-05 Module

The HC-05 is a wireless Bluetooth communication module used to transmit data between the Arduino UNO and the smartphone. It works on serial communication (UART) and enables wireless data transfer within a short range. In the Smart Wearable Glove project, the HC-05 module is connected to the TX and RX pins of the Arduino UNO.

After the Arduino recognizes a hand gesture using flex sensor data, it converts the gesture into a corresponding text message. This processed data is sent to the HC-05 Bluetooth module through serial communication. The Bluetooth module then wirelessly transmits the data to the paired Android smartphone or display device.

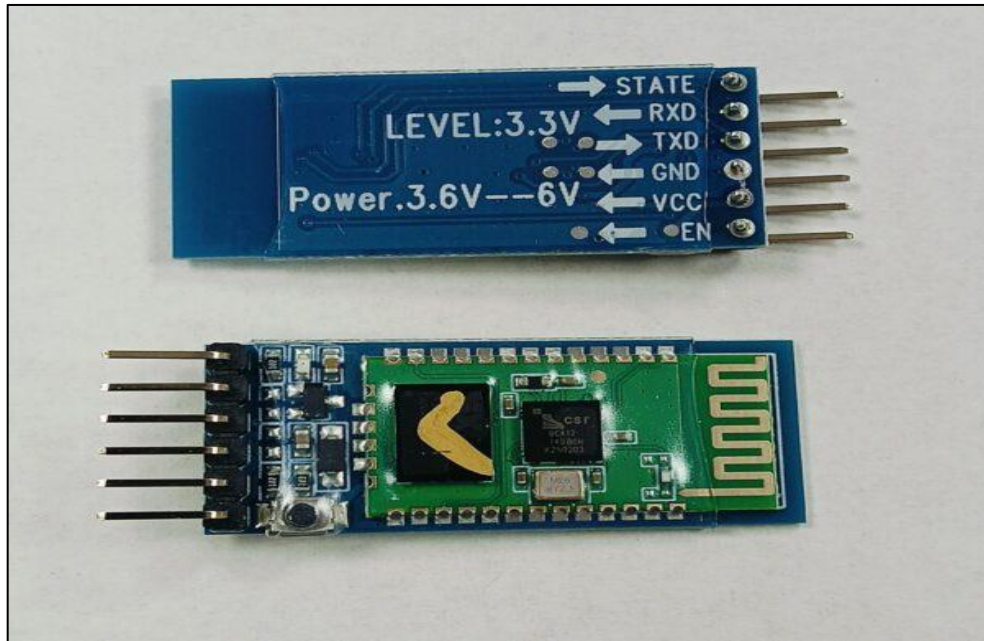
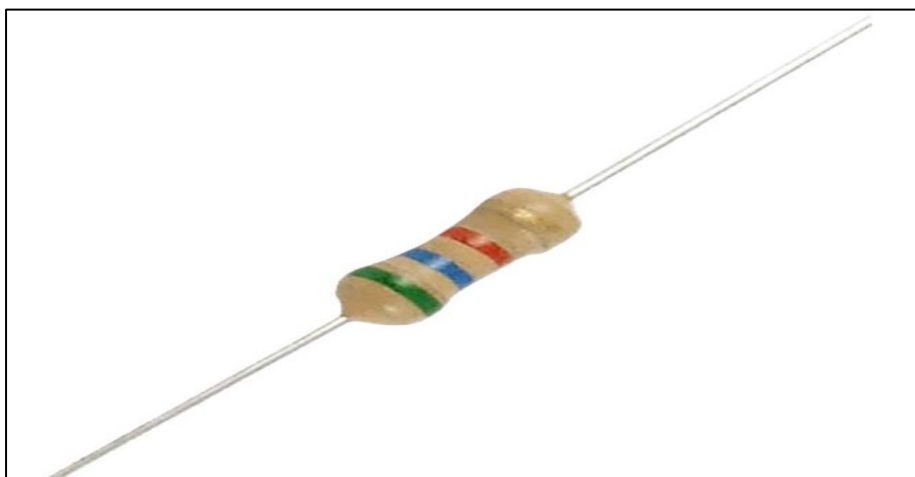


Figure5.2.3.1: Bluetooth HC-05 Module

The smartphone receives the transmitted data using a Bluetooth terminal application. The received message is displayed as readable text or converted into speech output for communication. Thus, the HC-05 Bluetooth module enables real-time wireless communication between the smart glove and the mobile device.

5.2.4. 10k Ω Resistors

The 10k Ω resistors are connected with flex sensors to form voltage divider circuits. Since Arduino can measure only voltage and not resistance directly, the resistor helps convert the changing resistance of the flex sensor into varying voltage signals. These voltage changes are then read accurately by the Arduino analog pins.

Figure5.2.4.1: 10k Ω Resistor

5.2.5 Wearable Glove

The wearable glove serves as the physical platform for mounting the flex sensors on the fingers. It ensures proper positioning of sensors for accurate detection of finger movements. The glove also provides flexibility, comfort, and easy handling during gesture performance.

5.2.6 Jumper Wires

Jumper wires are used to establish electrical connections between Arduino UNO, flex sensors, resistors, Bluetooth module, and power supply. They help in transmitting electrical signals and power throughout the system circuit.

5.2.7 Power Supply

The power supply provides the required 5V DC operating voltage to Arduino UNO and other electronic components. It ensures stable and continuous operation of the complete system during gesture detection and communication.

5.2.8 Android Smartphone

The Android smartphone acts as the output display device in the system. It receives the transmitted data from the HC-05 Bluetooth module through a Bluetooth application. The smartphone displays the translated text messages or converts them into speech output for communication with normal users.

5.3 Circuit Diagram

The circuit consists of multiple flex sensors, Arduino UNO, resistors, and the HC-05 Bluetooth module. Each flex sensor is connected in a voltage divider configuration using a resistor. One terminal of every flex sensor is connected to the 5V supply of the Arduino, while the other terminal is connected to both a resistor and an analog input pin of the Arduino UNO. The resistors are connected to the ground terminal, forming voltage divider circuits. Due to this arrangement, whenever the flex sensor bends, its resistance changes and produces different voltage levels at the analog input pins.

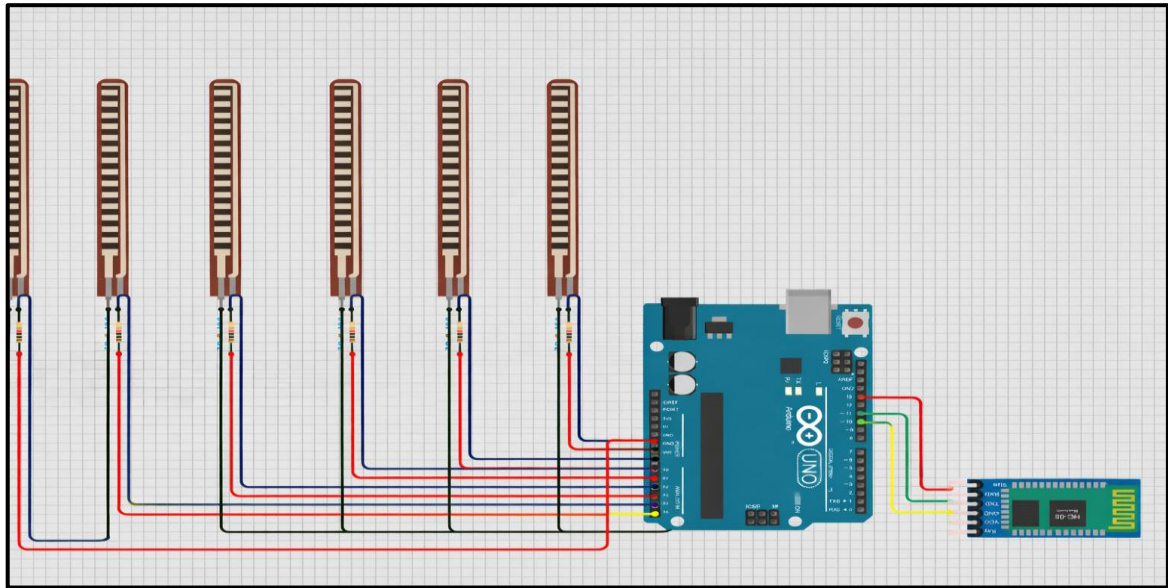


Figure5.3.1: Circuit Diagram of the Project

In the shown circuit, the flex sensors are connected to the analog pins A0, A1, A2, A3, and A4 of the Arduino UNO. These analog pins continuously read the varying voltage values produced by finger bending. The Arduino processes these sensor values and compares them with predefined gesture patterns stored in the program memory to identify different hand gestures.

The HC-05 Bluetooth module is connected to the digital communication pins of the Arduino UNO for wireless data transmission. The TX pin of the HC-05 is connected to the RX pin of the Arduino, and the RX pin of the HC-05 is connected to the TX pin of the Arduino. The VCC pin of the Bluetooth module is connected to the 5V supply, and the GND pin is connected to the Arduino ground. After gesture recognition, Arduino sends the corresponding text data to the HC-05 module, which wirelessly transmits the information to the Android smartphone for displaying the output message.

5.4 MATLAB Code

```
#include <SoftwareSerial.h>

SoftwareSerial BT(10, 11);

String lastMsg = "";

void sendMsg(String msg) {
    if (msg != lastMsg) {
        Serial.println(msg);
        BT.println(msg);
        lastMsg = msg;
    }
}

void setup() {
    Serial.begin(9600);
    BT.begin(9600);
}

void loop() {
    int f1 = analogRead(A0);
    int f2 = analogRead(A1);
    int f3 = analogRead(A2);
    int f4 = analogRead(A3);
    int f5 = analogRead(A4);

    Serial.print("A0: "); Serial.print(f1);
    Serial.print(" A1: "); Serial.print(f2);
    Serial.print(" A2: "); Serial.print(f3);
    Serial.print(" A3: "); Serial.print(f4);
    Serial.print(" A4: "); Serial.println(f5);
```

```
if (f1 < 750) {  
    sendMsg("I NEED FOOD");  
}  
else if (f2 < 750) {  
    sendMsg("I NEED WATER");  
}  
else if (f3 < 750) {  
    sendMsg("I NEED WASHROOM");  
}  
else if (f4 < 750) {  
    sendMsg("I NEED WALKING HELP");  
}  
else if (f5 < 750) {  
    sendMsg("I NEED SLEEP");  
}  
else {  
    sendMsg("NO RESPONSE");  
}  
  
delay(500);  
}
```

5.5 Developed Model



Figure5.5.1:Working Model of the Project

The above figure shows the working model of the Smart Wearable Glove for Sign Language Translation system developed using flex sensors, Arduino UNO, HC-05 Bluetooth module, resistors, and connecting wires. The flex sensors are mounted on the fingers of the glove so that they bend whenever the user performs hand gestures. When the fingers bend, the resistance of the flex sensors changes according to the bending angle. These resistance variations are converted into corresponding voltage signals using voltage divider circuits formed with resistors. The analog voltage outputs from the sensors are connected to the analog input pins of the Arduino UNO microcontroller.

The Arduino UNO continuously reads and processes the sensor values and compares them with predefined gesture patterns stored in the program memory. Once a gesture is recognized, the corresponding message or command is generated. The HC-05 Bluetooth module connected to the Arduino wirelessly transmits the recognized data to an Android smartphone or Bluetooth-enabled device. The received output is displayed as text or converted into speech, enabling real-time communication for deaf and mute individuals. The model demonstrates the practical application of embedded systems, sensor technology, and wireless communication in assistive communication systems.

CHAPTER 7

APPLICATIONS

7.1 Applications of this project

- 1. Communication Assistance:** The system helps deaf and mute people communicate easily using hand gestures. It converts sign language into understandable text or speech output.
- 2. Real-Time Sign Language Translation:** The glove translates hand gestures instantly without delay. This enables smooth and fast real-time communication.
- 3. Hospital Communication:** The device helps patients communicate with doctors and nurses easily. It is useful when interpreters are not available in hospitals.
- 4. Educational Institutions:** The system supports specially abled students during learning activities. It improves communication between teachers and students.
- 5. Smart Healthcare Devices:** The project can be used in wearable healthcare assistive systems. It improves interaction and patient support using smart technology.
- 6. Human-Computer Interaction:** The glove can be used for gesture-based control systems. It enables communication between humans and electronic devices.
- 7. Public Communication Support:** The system helps users communicate in public places independently. It reduces dependency on sign language interpreters.
- 8. Android Application Integration:** The glove can be connected to Android applications through Bluetooth. The mobile app displays translated messages or speech output.
- 9. Research and Development:** The project is useful for research in embedded systems and IoT. It helps in developing advanced gesture recognition technologies.
- 10. Smart Home Automation:** Hand gestures can be used to control smart home devices. The system can operate lights, fans, and appliances wirelessly.

7.2 Advantages

1. Helps deaf and mute people communicate easily using hand gestures.
2. Provides real-time translation of sign language into text or speech.
3. Enables wireless communication using the HC-05 Bluetooth module.
4. Portable and lightweight design makes it easy to use daily.
5. Low-cost components reduce overall system cost.
6. Reduces dependency on human interpreters for communication.
7. Simple and user-friendly system operation.
8. Consumes less electrical power during operation.
9. Can be upgraded using AI and machine learning techniques.
10. Improves social interaction and independence for specially abled individuals.

7.3 Disadvantages

1. Gesture recognition accuracy depends on proper sensor calibration.
2. The system can recognize only limited predefined gestures.
3. Bluetooth communication range is limited to short distances.
4. Flex sensors may lose sensitivity after long-term usage.

CHAPTER 8

FUTURE SCOPE

1. The proposed system can be upgraded using machine learning and artificial intelligence techniques for more accurate gesture recognition.
2. Additional sensors such as accelerometers and gyroscopes can be integrated to recognize complex hand movements and dynamic gestures.
3. Cloud and IoT technologies can be integrated for remote communication and data storage applications.
4. Wireless communication range can be improved using advanced communication modules such as Wi-Fi or GSM.
5. The glove can be integrated with smart home automation systems for gesture-based device control.
6. The project can be used in healthcare, education, and public communication systems for assisting specially abled individuals.

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